

Name : _____

Mark : _____/20 _____

Date: 09/11/2024

Period: 1 hour



REPUBLIC OF YEMEN
Science & Technology University
Aden Branch
Faculty of Engineering
Civil Engineer Department

CHEMISTRY TEST

For B. Engineering

2024-2025

Q1. (a) (4 marks) Write the electronic configuration using simplified notation and determine the number of protons, neutrons and electrons in; ${}_{20}^{42}\text{Ca}$

(b) (6marks) Give the increasing order (>) of the following:

- The radius: ${}_{19}^{39}\text{K}$, ${}_{6}^{13}\text{C}$, ${}_{7}^{14}\text{N}^{-3}$

- The mass: 10,00 μg , $10.00 \times 10^3 \text{ mg}$, 0.01 Kg, 1000,000 ng

Q2. (a) (3 marks) Determine whether each bond is ionic or polar/nonpolar covalent: C-O, N-H (EN: C=2.5, O=3.5, H=2.1, N=3.0)

(b) (3 marks) Calculate the number of moles and mass of each element in the following reaction if 1 mole of NaHCO_3 is used:



(A.wt. = H =1, Na=23, S=32, C =12, O=16)

Q3. (a) (2 marks) Perform the following unit conversion :
 $150^{\circ}\text{F} = ?\text{K}$

(b) (2 marks) Calculate and round the answer to two significant figures:
 $2.11 \times 23 / 2.4 - 3.001$